

Pretrained Machine Learning Models, Multimodal Sensor Fusion, and Benchmark Datasets for Automated E-Waste and Material Recovery

Regulatory Frameworks and Industrial Sorting Imperatives

The expansion of global waste electrical and electronic equipment (WEEE) presents severe environmental hazards alongside significant material recovery opportunities¹. In 2022, global e-waste generation reached approximately 62 million metric tons, with regions such as India producing over 4.1 million metric tons². In response, international regulatory bodies have established strict compliance regimes to mandate systematic recovery and hazardous component decontamination¹.

The European Union Waste Electrical and Electronic Equipment Directive (Directive 2012/19/EU) establishes an open-scope regulatory framework covering six main functional categories:

temperature exchange equipment; screens and monitors ($>100 \text{ cm}^2$); lamps; large equipment ($>50 \text{ cm}$); small equipment ($\leq 50 \text{ cm}$); and small IT/telecommunication equipment ($\leq 50 \text{ cm}$)⁴. The directive requires member states to achieve a minimum collection target of 65% of the average weight of electrical and electronic equipment (EEE) placed on the market over the preceding three years³.

Under Annex VII of Directive 2012/19/EU, facility operators must execute mandatory selective treatment to extract designated hazardous components prior to mechanical shredding or material recovery⁵. The mandatory removal list includes:

- Printed circuit boards (PCBs) assembled with electronic components possessing a surface area greater than 10 cm^2 ⁷.
- Components containing mercury, such as switches and backlights⁴.
- Plastics containing brominated flame retardants (BFRs)⁷.
- Asbestos waste and refractory ceramic fibers⁵.
- Chlorofluorocarbons (CFCs), hydrochlorofluorocarbons (HCFCs), hydrofluorocarbons (HFCs), and hydrocarbons (HCs)⁵.
- Electrolyte capacitors containing substances of concern⁵.
- Batteries, specifically lead-acid and lithium-based cells⁴.

Simultaneously, jurisdictions outside the European Union have enacted rigorous Extended Producer Responsibility (EPR) frameworks. Under India's Central Pollution Control Board

(CPCB) E-Waste (Management) Rules, the scope covers over 130 product categories². EEE producers face phased collection targets—set at 20% for periods extending through 2026—enforced by dual Environmental Compensation (EC) regimes². EC Regime 1 penalizes shortfalls in EPR collection and recycling targets based on average recovery costs, while EC Regime 2 enforces escalating financial penalties for operational non-compliance, unfulfilled quarterly returns, or improper disposal².

These regulatory mandates, coupled with the European Union Critical Raw Materials Act, have shifted industrial sorting from crude density separation toward automated, sensor-based robotic sorting¹. Automated systems require high-throughput vision and machine learning models capable of identifying chemical composition, component boundaries, and spatial trajectories under high line speeds⁹.

Current State of Pretrained Models in E-Waste and Material Recovery

Availability of Turnkey Foundation Models

A critical question in industrial computer vision is whether specialized, off-the-shelf "e-waste foundation models" exist in open-source repositories. Off-the-shelf, monolithic foundation models trained *exclusively* on end-to-end industrial e-waste streams do not exist as unified, ready-to-deploy weights.

Instead, the ecosystem relies on fine-tuning vision foundation models—such as DINOv2, Swin Transformer backbones, and YOLO variants (YOLOv8, YOLO11)—or transfer-learning from Earth Observation and Remote Sensing foundation models⁹. Specialized spectral architectures often utilize self-supervised pretraining strategies, such as Hyperspectral Masked Autoencoders (HyperMAE), to learn invariant spectral representations before fine-tuning on downstream waste sorting datasets¹⁵.

Earth Observation Foundation Models as Analogues

To process high-channel multispectral and hyperspectral imagery (ranging from Near-Infrared to Short-Wave Infrared), researchers frequently adapt pretrained weights from Earth Observation (EO) and Remote Sensing foundation models⁹. Systems such as Prithvi, Clay, and CROMA are pretrained on massive satellite repositories (e.g., Sentinel-2, Landsat) incorporating RGB, NIR, and SWIR bands¹⁶.

While EO models share channel overlap with waste sorting sensors, significant domain shifts limit their direct applicability:

- **Spatial Resolution:** Earth observation models operate on macro-scale spatial resolutions (10 to 30 meters per pixel), where spatial context is dominated by geography¹⁷. Industrial sorting requires sub-millimeter spatial resolution to detect small electronic components, microscopic circuit traces, or broken plastic shards⁸.
- **Spectral Characteristics:** Satellite sensors possess fixed, wide spectral bandwidths optimized for land cover, forestry, and atmosphere¹⁷. Waste recovery requires continuous, narrow-band hyperspectral imaging (400–2500 nm across hundreds of channels) to differentiate polymer chemistry (e.g., PET, PP, HDPE, LDPE, ABS, PS) and

metallic surfaces¹³.

- **Physical Condition:** Remote sensing images clear physical topography. Industrial e-waste streams contain heavily deformed, dirt-contaminated, partially occluded, and crushed objects moving on high-speed conveyor belts¹⁰.

Adapting General Vision Backbones via Domain Transfer

Due to these operational differences, state-of-the-art waste sorting pipelines employ domain adaptation on general vision backbones⁹. Models such as DINOv2 provide self-supervised visual features that transfer well to Long-Wave Infrared (LWIR) and thermal streams¹⁴.

Concurrently, real-time convolutional object detectors like YOLO11 utilize modified necks (such as lightweight GSConv modules and inverted Residual Mobile Block structures) to maintain high frame-per-second (FPS) rates while detecting tiny e-waste sub-components¹¹.

Multimodal Sensor Fusion and Model Architectures

Sensor Modalities and Technical Trade-offs

Single-modality vision systems face fundamental physical limitations when deployed in complex material recovery environments:

- **Visible RGB Imaging:** Captures high-spatial detail, surface texture, color, and geometric boundaries¹⁰. However, RGB cannot distinguish polymers of identical color or identify underlying flame-retardant chemistry⁷.
- **Hyperspectral Imaging (HSI - NIR/SWIR):** Operates across the 700–2500 nm range to measure molecular vibrational overtone absorption⁹. HSI differentiates chemical compositions (e.g., ABS vs. Polycarbonate) but suffers from low spatial resolution, low signal-to-noise ratios, and high computational dimensionality⁹.
- **Thermal / Long-Wave Infrared (LWIR):** Captures thermal dissipation signatures, aiding in material discrimination where surface active heating is applied, or during non-uniform illumination¹⁴.
- **Terahertz (THz) Imaging:** Penetrates non-metallic layers to reveal internal component structures and hidden metallic contacts beneath plastic casings²¹.

Deep Fusion Frameworks

To merge high-spatial RGB with high-spectral HSI without experiencing early spectral collapse or spatial degradation, deep learning fusion networks have been developed⁹.

Bidirectional Cross-Attention Fusion (BCAF)

BCAF addresses grid misalignment between high-resolution RGB (e.g., 1200×1184) and low-resolution HSI (e.g., 600×640)⁹. Rather than downsampling RGB or projecting HSI to pseudo-RGB channels via Principal Component Analysis (PCA), BCAF processes both modalities at their native grids⁹. It utilizes a dual-backbone setup: a standard Swin Transformer for spatial RGB context and an HSI-adapted Swin backbone with 3D tokenization and factorized spatial-spectral attention for the spectral domain⁹. Localized cross-attention layers align feature maps dynamically across scales, achieving real-time throughput⁹.

Multimodal Fusion Transformer (MFT)

Designed for complex waste composition (e.g., fuel cell and electrolyzer recycling), MFT combines spatial patch tokens from high-resolution RGB with spectral vector tokens from VNIR/SWIR (380–2500 nm) sensors¹⁹. MFT avoids mixed-spectrum token pollution during spatial convolutions by isolating spectral feature extraction prior to joint spatial-spectral attention²³.

Software-Based Multi-Sensor Registration

In industrial setups lacking hardware-synchronized frame grabbers, software-only spatial and temporal alignment is critical¹⁴. Algorithms utilize SuperPoint feature extraction combined with SuperGlue graph neural network matching to compute projective homographies between RGB and LWIR/HSI sensors¹⁰. Adaptive sliding-window temporal matchers compensate for belt speed variations and camera non-uniformity correction (NUC) events¹⁴.

Model Architecture	Input Modalities	Primary Benchmark	mIoU / Accuracy Performance	Throughput / Speed	Target Application
BCAF [cite: 9, 13]	High-Res RGB + Low-Res HSI	SpectralWaste	76.4% mIoU (31 FPS) / 75.4% mIoU (55 FPS)	31–55 FPS (RTX 4090)	Automated waste stream segmentation
BCAF [cite: 9, 13]	Native RGB + Multi-Slice HSI	K3I-Cycling	62.3% mIoU (Material) / 66.2% mIoU (Plastics)	Real-time industrial line speed	Sorting PET, PP, HDPE, LDPE, PS
COSNet [cite: 22]	Single RGB	ZeroWaste-f	56.67% mIoU	Standard inference speed	Boundary enhancement in cluttered waste
COSNet [cite: 22]	Single RGB	SpectralWaste	69.96% mIoU	Standard inference speed	Cardboard, film, rigid plastic

					segmentation
FANet [cite: 22]	Single RGB	SpectralWaste	67.83% mIoU	High-throughput baseline	Industrial stream segmentation
MFT (9 × 9) [cite: 19, 23]	High-Res RGB + HSI (380–2500 nm)	Electrolyzer s-HSI	69.30% Pixel OA / 98.16% Object OA	Off-line batch processing	Electrolyzer component recovery
Pixel-wise SpectralFormer [cite: 19, 23]	HSI Vector (380–2500 nm)	Electrolyzer s-HSI	59.78% Pixel OA / 98.02% Object OA	High-throughput vector processing	Pure spectral material identification

Benchmarks and Datasets for Pretraining and Fine-Tuning

In the absence of a single universal e-waste foundation model, training specialized architectures requires domain-specific benchmark datasets. These datasets cover two primary domains: printed circuit board defect/component segmentation and multimodal industrial waste streams.

Printed Circuit Board Datasets

PCBs contain high concentrations of critical raw materials alongside toxic elements, requiring accurate localization for Annex VII selective treatment⁷.

- **PKU-Market-PCB:** Derived from Peking University's public dataset and augmented for deep learning models¹⁸. Contains 3,505 high-resolution images categorized into six primary defect types: missing hole, mouse bite, open circuit, short, spur, and spurious copper¹⁸.
- **DeepPCB:** Contains 1,500 aligned image pairs consisting of a defect-free template and a tested image¹⁸. Annotations include normalized bounding boxes covering six defect classes (open, short, mousebite, spur, copper, pin-hole)¹⁸.
- **PCB-Vision:** A hybrid RGB-hyperspectral benchmark dataset comprising 53 multi-component scenes spanning 400–1000 nm (Specim FX10) paired with high-resolution RGB cameras¹⁹. Designed for sub-component material separation and

board-level recycling²⁵.

Complex Multimodal Waste Sorting Datasets

- **SpectralWaste:** A multimodal benchmark collected from an operational sorting facility¹⁹. Features 852 labeled scene pairs and 6,803 unlabeled scenes comprising synchronized RGB (1200×1184) and SWIR HSI (600×640 , 1000–1700 nm)⁹. Classes include cardboard, film, basket, filaments, video tape, and trash bags²².
- **Electrolyzers-HSI:** Designed for precious material recovery from green hydrogen equipment¹⁹. Contains 55 co-registered scenes featuring over 4.2 million pixel vectors (424,169 fully labeled) spanning 380–2500 nm across 450 spectral channels¹⁹. Target classes include titanium mesh, black steel, gray steel, HTEL anodes, and HTEL cathodes¹⁹.
- **ThermalRGBTrash:** A synchronized RGB-LWIR video benchmark containing 22 video pairs, 901 labeled tracklets across four material classes, and 1,003 RGB frames with 5,429 instance masks designed for thermal-assisted conveyor sorting¹⁴.
- **ZeroWaste-f:** A highly cluttered RGB dataset comprising 4,503 images (3,002 training, 572 validation, 929 testing) collected from material recovery facilities²². Covers four primary categories: cardboard, soft plastic, rigid plastic, and metal²².
- **RGB-HSI Municipal Solid Waste:** Focuses on second-stage flake sorting after mechanical crushing²⁰. Contains paired 640×640 RGB images and $640 \times 640 \times 224$ HSI data cubes (936–1722 nm) categorizing nine distinct polymer types (Color/Transparent/White PE, PP, and PET)²⁰.

Dataset	Modalities	Wavelength / Spectral Range	Spatial Resolution / Dimensions	Scale / Instances	Primary Task
PKU-Market-PCB [cite: 18]	Single RGB	Visible Spectrum	Enhanced Multi-Scale	3,505 Images	PCB Defect Detection
DeepPCB [cite: 18, 24]	Template + Test RGB Pairs	Visible Spectrum	Aligned Pixel Pairs	1,500 Image Pairs	PCB Micro-Defect Localization
PCB-Vision [cite: 19, 25]	RGB + HSI	400–1000 nm	Co-registered Scenes	53 Scenes	Electronic Component Segmentati

					on
SpectralWaste [cite: 9, 19, 22]	RGB + HSI (SWIR)	1000–1700 nm	RGB: 1200×118 HSI: 600×640	852 Labeled / 6,803 Unlabeled	Multimodal Industrial Waste Sorting
Electrolyzers-HSI [cite: 19, 23]	RGB + HSI (VNIR/SWIR)	380–2500 nm (450 Bands)	RGB: 4096 px line rate HSI: 384 px line rate	55 Data Cubes (4.2M Pixel Vectors)	Critical Material & Alloy Recovery
ThermalRGBTrash [cite: 14]	RGB + LWIR (Thermal)	Long-Wave Infrared	Co-registered Frames	22 Video Pairs (901 Labeled Tracklets)	Conveyor Waste Instance Segmentation
ZeroWaste-f [cite: 22]	Single RGB	Visible Spectrum	Variable Field-of-View	4,503 Images	Extremely Cluttered Stream Sorting
RGB-HSI MSW [cite: 20]	RGB + HSI	936–1722 nm (224 Bands)	640×640 : Tensor	Flake Multi-Class Annotations	Polymer Flake Classification

Real-Time Object Tracking on High-Speed Industrial Conveyors

Tracking-by-Detection Frameworks on Conveyors

Identifying material composition frame-by-frame is insufficient for robotic sorting arms or

air-jet ejectors. Systems must maintain persistent object identities (^{ID}) and spatial trajectories over time as items move on high-speed conveyor belts¹¹. Multi-Object Tracking (MOT)

frameworks follow either a Tracking-by-Detection (TBD) or Joint Detection and Tracking (JDT) paradigm¹¹. In industrial TBD pipelines, object tracking proceeds through four distinct operations:

Detection $\longrightarrow$ Motion Prediction (Kalman Filter) $\longrightarrow$ Cost Matrix Generation $\longrightarrow$ Data .

Algorithm Taxonomy and Mechanics

ByteTrack

Traditional trackers reject bounding boxes below a high confidence threshold (typically $\tau_{\text{high}} = 0.5$), discarding occluded, blurred, or dirty items²⁶. ByteTrack solves this using a two-stage association pipeline²⁶.

In Stage 1, high-confidence detections ($\tau \geq \tau_{\text{high}}$) are associated with active tracks using Intersection over Union (IoU) or appearance distance²⁶.

In Stage 2, remaining unmatched tracks are associated against low-confidence detections ($\tau_{\text{low}} \leq \tau < \tau_{\text{high}}$) using strict IoU filtering²⁷. This recovers occluded or distorted e-waste items without introducing false positives²⁶.

BoT-SORT

Combines motion and visual appearance features by integrating Global Motion Compensation (GMC) into the Kalman filter to account for camera jitter or belt speed variations²⁶. It incorporates deep Re-Identification (ReID) feature embeddings extracted via backbones like FastReID (e.g., DukeMTMC ResNet-50 models) directly into the linear assignment cost matrix²⁶.

OC-SORT (Observation-Centric SORT)

Standard Kalman filters assume constant velocity and linear motion²⁶. When e-waste items bounce, collide, or undergo temporary occlusion on a conveyor, the filter state variance expands, causing trajectory drift upon re-emergence³⁰. OC-SORT corrects this drift by running a virtual trajectory re-update between the last known observation and the new observation, substituting observation-centric velocity direction for noise-dominated filter estimates³⁰.

DeepSORT and LITE Variants

DeepSORT serves as the legacy baseline, employing a standard 8-dimensional state vector Kalman filter combined with deep appearance features²⁶. However, traditional DeepSORT exhibits high computational overhead during feature extraction, dropping inference speeds to 8–10 FPS²⁹.

The LITE framework (e.g., LITE-DeepSORT, LITE-BoT-SORT) decouples and optimizes feature extraction, utilizing intermediate YOLO layer representations (e.g., layer 14 of YOLOv8m) to achieve 25–38 FPS without compromising tracking accuracy²⁹.

Tracki ng	Assoc iation	Motio n	ReID Appe	Bench mark	HOTA	MOTA	Throu ghput	Prima ry
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Algori thm	Meth odolo gy	Model	aranc e Model	(Data set)	(%)	(%)	(FPS)	Opera tional Suitab ility
ByteT rack [cite: 26, 27, 28, 29]	Two-S tage High/L ow Confid ence IoU	Linear Kalma n Filter	None (IoU Only)	MOT1 7 Validat ion	60.9	75.7	High (>50 FPS)	High-t hroug hput, dense stream s with low comp ute
BoT-S ORT [cite: 11, 26, 29]	High- Confid ence + GMC + ReID	GMC- Kalma n Filter	FastRe ID (ResN et50)	MOT1 7 Test	40.8	Base Ref.	10.6 FPS	High-a ccurac y sortin g under camer a move ment
LITE- BoT-S ORT [cite: 29]	Optimi zed LITE ReID Featur e Coupli ng	GMC- Kalma n Filter	Layer- 14 Featur e Extrac t	MOT1 7 / MOT2 0	40.8 / 21.1	Bench marke d	38.2 FPS / 31.8 FPS	Real-ti me indust rial line sortin g
OC-S ORT [cite: 26, 30]	Obser vation -Centr ic Traject	Virtual Obser vation Filter	Optio nal	Indust rial / MOT	SOTA Ref.	Superi or Occl.	High FPS	Non-li near item motio n,

	ory Re-up date							tumbli ng, collisio ns
Deep SORT [cite: 26, 29, 31]	Stand ard Casca ded Matchi ng	Stand ard Kalma n Filter	Deep ReID CNN	MOT1 7 / MOT2 0	43.7 / 24.4	Baseli ne	10.5 FPS / 8.5 FPS	Legac y baseli ne refere nce
LITE- Deep SORT [cite: 29]	Decou pled Lightw eight ReID	Stand ard Kalma n Filter	Layer- 14 Interm ediate	MOT1 7 / MOT2 0	43.0 / 25.2	High Recov.	26.7 FPS / 15.9 FPS	Embe dded deploy ment on roboti c sortin g cells
IEGS- BoT [cite: 11]	Enhan ced Cell/O bject Tracke r	Invert ed Residu al / GSCo nv	Integr ated Detect or	Compl ex Micros copic	51.3	53.97	Real-Ti me	Small, defor med e-was te comp onent trackin g

Synthesis and System Integration Roadmap

Achieving fully automated e-waste recovery requires integrating regulatory requirements, multimodal sensor hardware, specialized neural architectures, and real-time tracking algorithms³. Rather than relying on a single turnkey model, state-of-the-art deployments utilize a modular five-stage processing pipeline:

1. **Hardware Ingestion Layer:** High-speed conveyor belts pass material under synchronized high-resolution RGB cameras (e.g., 1200×1184 at 16 KHz line rate),

hyperspectral sensors (spanning the 380–2500 nm VNIR/SWIR spectrum), and thermal LWIR sensors⁹.

2. **Registration & Alignment Layer:** In the absence of hardware-locked frame grabbers, software alignment uses SuperPoint feature detection coupled with SuperGlue graph neural networks¹⁴. Adaptive sliding-window temporal matchers compute projective homographies to warp multi-sensor frames into a unified coordinate system¹⁴.
3. **Multimodal Feature Fusion Layer:** Deep fusion architectures process co-registered streams⁹. Models such as Bidirectional Cross-Attention Fusion (BCAF) maintain native RGB spatial resolution while extracting 3D spectral tokens from HSI slices, avoiding early spectral collapse or downsampling artifacts⁹.
4. **Trajectory & Tracking Layer:** Multi-object tracking algorithms process fused detection frames to maintain persistent object identifiers¹¹. Algorithms such as ByteTrack or LITE-BoT-SORT evaluate high- and low-confidence bounding boxes, updating state trajectories via Kalman filtering at rates exceeding 30 FPS²⁷.
5. **Action & Execution Layer:** Object positions and trajectory vector estimates are passed to physical actuators¹⁰. High-speed pneumatic air-jet manifolds eject specific plastic polymers, while multi-axis robotic pick-and-place systems remove PCBs and components requiring mandatory Annex VII treatment⁵.

Key Strategic Takeaways

The transition toward fully automated e-waste sorting relies on three critical advancements:

- **Absence of Turnkey Foundation Models:** Monolithic, end-to-end e-waste foundation models do not currently exist in open-source registries. Current industrial architectures depend on domain transfer from general vision backbones (DINOv2, Swin, YOLO11) or adaptation of Earth Observation models (Prithvi, Clay) fine-tuned on specialized waste datasets¹¹.
- **Self-Supervised Spectral Pretraining:** Self-supervised pretraining strategies, specifically Hyperspectral Masked Autoencoders (HyperMAE) with decoupled spatial-spectral masking, provide the most effective mechanism for pretraining on unlabelled hyperspectral waste streams prior to downstream fine-tuning¹⁵.
- **High-Throughput Tracking & Fusion:** Achieving industrial viability requires balancing fusion accuracy with inference latency. Architectures like BCAF paired with lightweight trackers like LITE-BoT-SORT or ByteTrack enable real-time processing (> 30 FPS), allowing sorting facilities to satisfy strict collection and recovery mandates established by international e-waste regulations².

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